Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 1 of 11 · Question 7 part a.ii · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a)	(i)	• 2nd set of parents have genotypes I^BI^O and I^BI^O/both parents are I^BI^O; • give rise to child I^OI^O which is M; • 1st set of parents have genotypes I^OI^O and I^AI^B; • can give rise to a child I^AI^O which is N;	Allow 1 mark if the child is matched to the correct parent.	4
	(ii)	• parental genotypes $I^BI^O \times I^BI^O$; • offsprings are I^BI^B 2 I^BI^O I^OI^O; • 75%/0.75/3:1/3 in 4;	accept parental gametes for mp1 if BO and BO parental genotypes are used they can get ecf for MP2. accept offsprings in genetic cross	3
(b)		• (prick finger) with pin/needle; • sterile; • (smear) blood on slide; • thinly; • dry; • stain;	accept example of sterile Accept MP5 and MP6 in either order	5
				Total =12

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 1 of 11 · Question 7 part a.ii · 3 marks

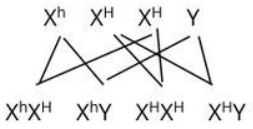
2025 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 2 of 11 · Question 9 part c.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  ;	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 2 of 11 · Question 9 part c.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 3 of 11 · Question 6 part b.iii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
6 (a)	(i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
	(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
(b)	(i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
	(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
	(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
	(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 3 of 11 · Question 6 part b.iii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 1 of 7 · find part a.iv in this mark scheme extract

4 (a) (i)

Africa

Blood group	Percentage of blood group
O	68
A	17/17.5
B	10;
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	40
B	12
AB	3;

2

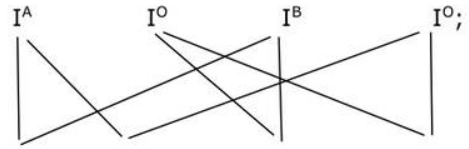
(ii)

- reference to antigens on red blood cells (group B);
- antibodies produced;
- causing agglutination/clumping (of red blood cells);

3

(iii)

I^AI^O I^BI^O;



I^AI^B I^AI^O I^BI^O I^OI^O;

Full marks for a Punnett square if parent genotypes are given.

3

(iv) 1 in 4:/25%/¼/0.25;

1

(b)

- <u>deoxygenated</u> blood moves into the left side/ventricle of the heart;
- (deoxygenated blood) mixes with <u>oxygenated</u> blood;
- less oxygen travelling around the body/to body/cells/in bloodstream;
- breathes faster to get more oxygen into the body;

4

Total 13

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 2 of 7 · find part a.iv in this mark scheme extract

	Answer	Notes	Marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 3 of 7 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks
6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks
7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 4 of 7 · find part a.iv in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 5 of 7 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 6 of 7 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks
9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 4 of 11 · Question 4 part a.iv · 1 mark

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 5 of 11 · Question 6 part b.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

6	(a) (i)	• prevent excess blood loss; • prevent entry of pathogens/bacteria/microorganisms;		2
	(ii)	V = platelets; W = thrombokinase enzyme; Y = fibrinogen; Z = fibrin;		4
	(b) (i)	• carried on X chromosome; • male must be X^hY; • female must be X^HX^h; • as allele is recessive; • but appears in next generation; any four from		5
	(ii)	• female has X^hX^h genotype; • so male offspring must be X^hY; • probability of a boy is 0.5; • probability of haemophilia is 1.0; • so probability of a boy with haemophilia is 1.0 × 0.5 = 0.5		4

Total 15 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 5 of 11 · Question 6 part b.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

8(a)(i)	broken down in liver; by deamination; to form urea; used to make new protein/amino acids;		**(3)**

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	poor/restricted movement of synovial joints pain in joint; backflow of blood in heart; poor/inefficient blood circulation/oxygenation; less energy/tired;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
8(b)(i)	• not sex-linked; • more females affected than males/more males would be affected; • only females would be carriers (if sex-linked); • male 1/3/11 would show condition if sex-linked; • but these males are carriers;		**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	• probability that child has condition is ¼ /1:3 / 1 in 4/25%/0.25; • probability that child is male is ½/1 in 2/1:1 / 50%/0.5; • so probability that it is affected male is 12.5%/0.125/$\frac{1}{8}$;		**(3)**

Total 13 Marks

Paper Total 90 Marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 6 of 11 · Question 8 part b.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 7 of 11 · Question 8 part b.ii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

8 (a)	any three from • change in base sequence/named mutation (1) • of DNA (1) • causing change in genotype/gene (1) • incorrect protein produced (1) • change in phenotype/characteristics (1)	Allow change in DNA for mps 1 and 2 (1)	3
(b) (i)	father X^HY mother X^HX^h (1) gametes X^H Y X^H X^h (1) F1 X^HX^H normal female; X^HX^h carrier female; X^HY normal male; X^hY haemophiliac male (1)		4
(ii)	(chance of first and second child is) ¼/0.25/25% (1) for both is ¼ x ¼/0.25 x 0.25 (1) = 1/16/0.0625/6.25% (1)		3
(c)	any five from • males only have one X chromosome (1) • two X chromosomes/H and h needed to be a carrier (1) • recessive allele/h (1) • carried on X chromosome (1) • either present or not (1) • no chance for dominant allele to be present (1) • to mask recessive (1)		5

Total 15 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 7 of 11 · Question 8 part b.ii · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 8 of 11 · Question 8 part b.ii · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
8 (a)	(i)	absorbs uv light; reduces/prevents risk of skin cancer;	Allow protects against (UV) radiation	2
	(ii)	eye/ retina/ choroid/ red blood cell/ haemoglobin ;		1
(b)	(i)	mother x father Hh hh; gametes H h h; fertilisation Hh hh; phenotype white forelock normal;	Reject diagrams that show sex linkage	4
	(ii)	chance of producing a boy is 0.5/½; chance of producing offspring with condition 0.5/½; chance of producing boy with condition is ½ × ½ = ¼;	Allow 50% or equivalent anywhere in calculation Full marks for correct final answer	3
(c)		Any 4 from different (genetic) code produced; different order of amino acids; codes for different proteins/enzymes; enzyme substrate complex not formed; causes change in pigment;	Allow change in base sequence/genetic code Allow any reference to change in amino acids	4

Total 14 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 8 of 11 · Question 8 part b.ii · 3 marks

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 9 of 11 · Question 7 part b.i · 3 marks

2022 JAN · Paper 02 · find part b.i in this mark scheme extract

number				
7 (a)	(i)	• appears in both males and females; • in similar numbers;		2
	(ii)	• condition present in heterozygous genotype/only one allele needed/affected are carriers; • only some children (from affected and unaffected parents) are affected/not affected;		2
(b)	(i)	• probability of a boy is 0.5/50%; • probability of no Huntington's is 1.0; • overall probability is 0.5 × 1.0 = 0.5/50%/1;		3
	(ii)	person 3 must be hh/homozygous recessive; does not carry a faulty allele; person 2 is Hh/heterozygous/carries a faulty allele / carries one normal allele; as father not affected; offspring either Hh or hh; half with condition and half without/50% chance of being affected;	marks can be awarded for suitable genetic crosses that are annotated	6

Total 13 marks

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Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 10 of 11 · Question 5 part c.ii · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

5 (a) (i)	• alternative form of gene; • only expressed in homozygote/when 2 recessive alleles are present/ when no dominant allele is present/not expressed in heterozygote/;			2
(b)	any two from • person 2; • person 4; • person 9;			
(ii)	Aa;			1
(c) (i)	parents Aa × Aa; gametes A a A a; F1 AA 2Aa aa; phenotype normal normal carrier sickle-cell;	2 marks max for unlabelled Punnett square (mp's 2 and 3)		4
(ii)	• chance of a boy = 0.5/50%; • chance of sickle cell anaemia = 0.25/25%; • chance of boy with condition = 0.5 × 0.25 OR $\frac{50 \times 25}{100}$; • = 0.125/12.5%;			4

Total 13 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 10 of 11 · Question 5 part c.ii · 4 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 11 of 11 · Question 5 part b.iii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

5 (a)	(i)	• change in nucleotide/base/codon sequence; • of DNA;		2
	(ii)	• new/different (sequence of) amino acids; • different protein / enzyme / different polypeptide • different phenotype/enzyme does not bind to substrate;		3
(b)	(i)	• allele for haemophilia on X chromosome/ not on Y chromosomes; • hairy ears could only be males / haemophilia could occur in females (and males);		2
	(ii)	• 1 = XX; • 2 = XY^H;		2
	(iii)	• chance of being a girl = 50%/ this can be indicated in a punnet square; • girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X; • no Y chromosome so no chance of hairy ears;		3

Total 12 marks

Type MS · Work out the probability of a child's genotype or phenotype from a cross

Answer 11 of 11 · Question 5 part b.iii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks